

ISO 22301 Graceful Degradation Architecture & Auditor Defence Brief

Platform: Community Emergency Action Platform (Community Hub)

Standard Reference: ISO 22301:2019 Clauses 7.5 (Documented Information), 8.1 (Operational Control), 8.4 (Business Continuity Procedures), and 8.4.4 (Mitigation and Response)

Core Technologies: Google Firebase (Hosting, Multi-Region Firestore), Git Version Control, Google Antigravity (Governed Agentic IDE)

Classification: Operational Architecture & Compliance Defence Evidence

1. Executive Summary for Auditors & Civil Resilience Teams

A recurring challenge during municipal resilience audits and emergency planning reviews is:

"If cellular masts collapse and internet backbones fail, how does a digital web app provide continuity? Furthermore, how do you ensure rapid AI-driven code changes never compromise life-safety systems?"

Under **ISO 22301**, organizations are not required to defy physical disruptions to public utilities. Rather, they are required to demonstrate:

1. **Pre-planned, tested fallback procedures (Graceful Degradation)** down to manual contingencies.
2. **Immutable change control and codebase continuity** so the platform itself can be restored, reviewed, or rolled back instantly.

Community Hub accomplishes this through a **4-Tier Continuity Model** supported by an auditable **Firebase + Git + Antigravity** technical spine:

[Tier 1: Full Cloud Connection] —► Firebase Multi-Region Edge, Web Sockets, Live GPS
| (Telecom masts congest/fail)



[Tier 2: Offline Client Memory] —► Firebase Persistent Disk Cache + Indexed DB (Local reads)
| (Wide-area ISP severed completely)



[Tier 3: Low-Bandwidth Fallback] —► 2G / SMS Gateways & Evacuation Shelter Subnet
| (Extended blackout / flat batteries)



[Tier 4: Manual "Grab-Bag" SOP] —► Pre-printed, waterproof quarterly physical PDF dossiers

2. Technical Breakdown Across Continuity Tiers

Tier 1: Normal Operations (Firebase Multi-Region Cloud)

- **Hosting & Edge Delivery:** Firebase Hosting backed by Google's global Edge CDN, serving pre-compressed assets from geo-distributed points of presence.
- **Database Layer:** Cloud Firestore deployed in **Multi-Region mode** (e.g., nam5 or eur3), providing automatic replication across physically separate metropolitan zones with zero-downtime failover.
- **Point-in-Time Recovery (PITR):** Continuous change-log retention enables microsecond-level state recovery up to 7 days if data corruption or unauthorised overwrites occur.

Tier 2: Degraded Network / Tower Loss (Offline-First Firebase PWA)

- **Mechanism:** Firebase Firestore Web SDK persistent Local Cache combined with custom Service Worker caching.
- **Auditable Evidence:**
 - Emergency Action Plans (EAPs), shelter matrices, lockbox combinations, and warden call-trees are automatically written to local client storage (Indexed DB) on initial retrieval.
 - When navigator.onLine === false, the application switches seamlessly to read from device flash storage without throwing unhandled exceptions or presenting blank screens.
 - Outbound field updates (e.g., road blockage reports) are buffered into an encrypted local outbox queue and automatically replayed with timestamp preservation once connectivity resumes.

Tier 3: Low-Bandwidth & Islanded Network Redundancy

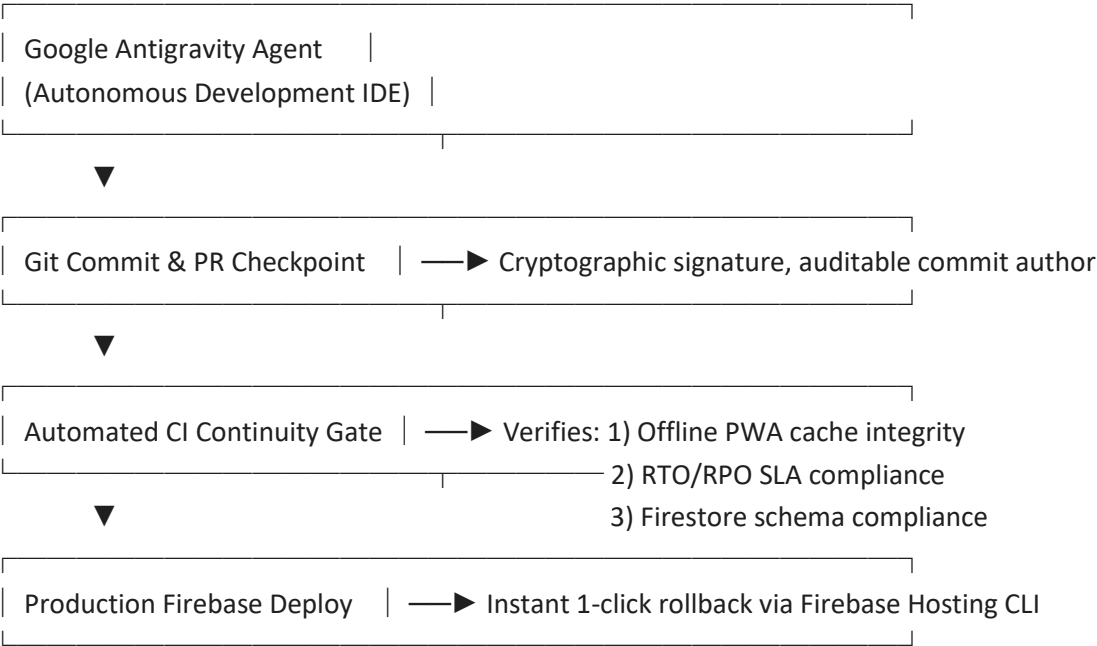
- **SMS / 2G Cellular Bridge:**
 - High-payload UI assets (maps, satellite tiles) are bypassed.
 - Incident notifications route through lightweight SMS gateways (e.g., GovNotify or Twilio) over 2G cellular frequencies that benefit from extended battery-backup survival at base stations.
- **Evacuation Shelter Micro-Hub:**
 - If a parish hall or evacuation shelter operates an isolated generator-powered Wi-Fi router without wide-area internet backhaul, clients connect locally to query cached peer data.

Tier 4: Non-Digital Contingency ("Grab-Bag" Protocol)

- **Standard Alignment:** ISO 22301 Clause 8.4 mandates manual workarounds when technology infrastructure outage limits are exceeded.
- **Implementation:**
 - An automated, client-side **"Export Resilience Dossier"** generates a high-contrast, 2-page print-optimised PDF.
 - Parish coordinators are instructed to print hard copies quarterly and place them into physical emergency grab-bags and hall lockboxes.
 - The physical dossier records analogue backup radio channels (PMR446 / VHF), generator cold-start SOPs, AED access keys, and vulnerable resident registers.

3. Engineering Governance & Change Control: Git & Google Antigravity

Auditors frequently inspect the software supply chain to verify that emergency tools cannot be taken down by faulty deployments or unverified AI agent contributions.



A. Git Version Control as Disaster Recovery (ISO 22301 Clause 7.5 & 8.1)

1. **Cryptographic Audit Trail:** Every configuration change, schema migration, and emergency protocol update is tracked via signed Git commits, ensuring absolute traceability of who approved each change.
2. **Deterministic Infrastructure-as-Code (IaC):** Firebase security rules, Firestore indexing definitions, and deployment targets are committed to the repository. The entire

production stack can be rebuilt from scratch in another cloud project in minutes.

- 3. **Instant Reversion (RTO Minute):** If an errant update impacts field operations, Firebase Hosting combined with Git tags allows a 1-command rollback to the last verified release without rebuilding containers.

B. Google Antigravity Agentic Governance (ISO 22301 Clause 8.3 & 7.2)

- 1. **Agentic Verification Gates:** Because code modifications and emergency workflows can be authored by AI agents within Google Antigravity, all agent commits must pass continuous integration (CI) test suites before hitting main.
- 2. **Offline-Mode Non-Regression Testing:** Antigravity automated test scripts deliberately sever network mock connections (offline event emission) to verify that recent code iterations have not broken the Indexed DB cache or Service Worker precaching manifests.
- 3. **Restricted Agentic Privileges:** Antigravity agents cannot deploy directly to live production environments; deployments require explicit authenticated human sign-off via branch protection rules.

4. Auditor Q&A Defence Script

Use these precise technical defences when responding to questions from civil protection boards, insurance underwriters, or ISO certification assessors:

Auditor Question	Compliant ISO 22301 Technical Answer
<i>"What happens to citizens when cellular towers drop?"</i>	"The application utilises client-side Service Workers and Firebase Indexed DB caching. Any resident who has previously accessed their local plan retains complete read access to maps, shelter lists, and warden contacts, even in complete airplane mode."
<i>"How do you prevent data loss if coordinators enter reports while offline?"</i>	"Submissions write immediately to an IndexedDB local outbox queue. Once any wireless handshake or cellular link re-establishes, records synchronise automatically to Cloud Firestore using Point-in-Time logs to prevent collision."
<i>"What if local power is out for 5 days and mobile batteries die?"</i>	"Conforming to ISO 22301 non-digital fallback mandates, the system includes a 1-click automated grab-bag exporter. Coordinators keep physical, laminated 2-page dossiers in parish hall safes containing

	radio channels and access PINs."
"What if Google Firebase suffers an outage?"	"Our database operates on Cloud Firestore Multi-Region infrastructure, spanning independent data centre zones with automatic failover. Additionally, the PWA client continues functioning on cached local device storage completely independent of cloud uptime."
"How do you govern AI agents in Google AntigraVity so they don't break emergency continuity?"	"All code generated in Google AntigraVity is bounded by strict Git version control and CI test suites. Automated tests simulate disconnected network environments to prove offline persistence before any deployment is accepted into production."

5. Public-Facing App Disclaimer & Terms Wording

(Place this text in your app's Terms of Service, registration flow, or Settings > About screen)

Emergency Service Continuity & Limitation of Service

1. **Intended Role:** The Community Hub platform is a decentralised community resilience and coordination tool. It is designed to assist local neighbourhood preparedness, volunteer organisation, and mutual aid. It is NOT a replacement for statutory emergency blue-light services (e.g., 999, 911, or 112). In any life-threatening situation, always attempt to contact official national emergency services first.

2. **Telecommunications Dependence & Local Caching:** While this platform incorporates advanced offline persistence (Indexed DB, Service Worker caching, and multi-region cloud infrastructure) to maintain availability during disruptions, live telecommunications rely on third-party mobile operators and public utility grids.

3. **Grab-Bag Hard-Copy Protocol:** In accordance with ISO 22301 community resilience guidelines, community wardens and coordinators are strongly advised to generate, print, and securely store physical hard copies of their emergency action dossier on a quarterly basis to guarantee operational readiness during total electrical or communications blackouts.

Layer 1: It Lives in the Leader’s Device Memory (Indexed DB Local Cache)

Whenever the leader or coordinator opens Community Hub during normal times:

- The app automatically downloads and saves a copy of the emergency plan, shelter contacts, keyholder numbers, and generator keys directly into the device's **local browser database (Indexed DB)**.
- **When the network collapses:** The app doesn't ask the cloud for anything. It opens instantly from the phone or laptop’s internal hard drive and generates the Grab-Bag dossier on the screen **even in Airplane Mode with zero WiFi and zero cellular signal**.

Layer 2: The Physical "Pre-Printed" Grab-Bag Protocol (ISO 22301 Gold Standard)

Under ISO 22301, true disaster recovery never relies *exclusively* on a digital screen during a storm (because phone batteries can die, screens can break, or laptops can run out of power after 48 hours).

The protocol is:

1. **Quarterly Routine (During Peacetime):** Every 3 or 6 months, the community resilience leader opens the app, clicks "**Export Grab-Bag Dossier (PDF)**", and prints **2 physical paper copies**.
2. **Lockbox Storage:** One copy goes into the **Village Hall emergency lockbox** and one goes into the **Leader's physical emergency backpack/grab-bag** alongside battery torches, generator keys, and analogue walkie-talkies.

Summary

Scenario	How the Grab-Bag Works
Normal Operations	Cloud syncs real-time changes instantly.
Internet & Cell Towers Down	Device opens the cached Grab-Bag directly from phone/laptop local memory (Indexed DB).
Power Grid Out & Batteries Dead (Day 4+)	Wardens pull the pre-printed, physical hard-copy dossier from the village hall lockbox.

This multi-layer redundancy is what guarantees **zero single points of failure**.